

2. Feature Engineering & Feature Selection

2.1 Economic and business rationale

Every material feature should make sense.

For a PD model, plausible drivers could include creditworthiness, LTV, indebtedness, loan characteristics and relevant economic factors.

For LGD, important drivers might include LTV, property value, geography, recovery characteristics and collateral-related factors.

Ask:

Why should this variable affect default/loss?

Then compare the theoretical direction with the observed relationship.

If higher LTV unexpectedly produces lower predicted LGD, that deserves investigation even if statistically significant.

2.2 Feature stability

A variable may be predictive during development but unstable over time.

Examine distributions and relationships across:

- years;
- vintages;
- geographic regions;
- economic regimes;
- lender groups;
- relevant portfolio segments.

You can use distribution comparisons, PSI/CSI where appropriate, missingness trends, category-frequency changes and changes in the relationship with the target.

The important question is whether the feature represents a **persistent risk relationship or merely a development-sample artifact**.

2.3 Variable transformations

Review every material transformation:

Raw variable → transformed feature

Examples include logarithms, bins, WOE, scaling, normalization, caps/floors, splines, interaction terms and derived ratios.

For each transformation ask:

- Why was it needed?
- Is it mathematically correct?
- Is it economically reasonable?
- Was the transformation selected using only training information?
- Does it improve stability or merely improve in-sample fit?
- Can it be reproduced in production?

For mortgage models, LTV is itself a derived feature, so even apparently straightforward variables can require substantial validation.

2.4 Multicollinearity and redundancy

Highly related features may carry essentially the same information.

Examples might include:

Original mortgage amount / outstanding balance / property value / LTV

Depending on methodology, assess correlation matrices, VIF and other dependency measures.

The issue isn't simply whether correlation exists. Determine whether it creates:

- unstable coefficients;
- counterintuitive signs;
- excessive sensitivity;
- difficulty interpreting the model;
- redundant complexity.

For tree-based ML models, traditional VIF is less relevant, but redundant correlated features can still affect interpretation and feature importance.

2.5 Target/data leakage

This is critical.

A feature must not contain information that would only become known **after the prediction point**.

Suppose the PD model predicts default at mortgage origination but a feature incorporates subsequent delinquency information. Performance could look excellent because the model is effectively seeing the future.

For every feature ask:

Was this information genuinely available at the exact time the model would have generated this prediction?

This should be tested both conceptually and through data timestamps.

2.6 Categorical variables and unseen categories

Review how variables such as province, property type, lender category and other categorical fields are encoded.

Check:

- rare categories;
- grouping methodology;
- missing categories;
- unknown categories;
- categories appearing only in validation/production;
- production fallback treatment.

The model should behave predictably when a category appears that wasn't present during training.

2.7 Feature availability at model-use time

This is related to leakage but broader.

A feature might not technically contain future information but may still not be reliably available operationally.

Ask for every important feature:

Is it available? At what point? From what system? At what quality? At what latency?

Development models sometimes use variables that cannot be consistently populated in production.

2.8 Feature sensitivity

Change important features while holding others constant and observe predictions.

For example:

LTV 70% → 80% → 90% → 100%

Then observe PD/LGD.

Do the same for important risk drivers.

You're looking for:

- economically sensible direction;
 - reasonable magnitude;
 - smooth/reasonable behaviour;
 - unexplained jumps;
 - excessive sensitivity;
 - unexpected insensitivity.
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2.9 Proxy and fairness concerns

A variable may be predictive because it indirectly represents another characteristic.

Geography is a classic example.

Assess whether selected features create unwanted bias or unfair outcomes, especially where proxies for sensitive characteristics may exist.

This doesn't mean automatically removing every correlated variable. It means understanding the relationship, purpose, impact and controls.

2.10 Feature-selection methodology

Finally, validate **how the final variables were chosen**.

Review:

- original candidate universe;
- screening criteria;
- statistical criteria;
- economic/business rationale;
- variable removals;
- stability criteria;
- multicollinearity considerations;
- expert judgment;
- final selection process.